

SPORTS COMPLEX MANAGEMENT SYSTEM (SCMS)

A digital platform to replace manual ID-card-based equipment issuing
in college sports complexes

Presented By :- Harshjot Singh • Jaspreet Singh • Sanveer Singh

B.E — Artificial Intelligence &
Machine Learning

Software Engineering project - 2026



PROBLEM STATEMENT

Challenges in the Existing System

- **Manual & inefficient process** – Students currently deposit physical ID cards to borrow sports equipment — slow, unrecorded, hard to track
- **No centralized record of inventory** (what's issued, available, or damaged)
- **No fair mechanism** to prevent a few students from monopolizing high-demand equipment/slots
- **Need:** an online system for issuing, tracking, and fairly scheduling equipment usage

PROPOSED SOLUTION

Our Proposed System

- **Digital** ID-based issue/return via QR code scan (no physical card handover)
- **Real-time inventory** dashboard for sports complex staff
- **Fairness**-weighted scheduling algorithm — prioritize students who've played less recently, capped daily usage
- **Extras:** usage analytics, return reminders, damage logging



MOTIVATION

Why SCMS?

- **Real-world impact** – Current process is manual, paper/register-based, and prone to errors
- **Application of Software Engineering** – Provides practical experience in requirements analysis, system design, database modeling, implementation, and testing.
- **A digital** platform to replace manual ID-card-based equipment issuing in college sports complexes
- **Better campus resource management** – Improves student convenience while increasing equipment accountability, utilization, and system efficiency.